

Sai Gopal Reddy Kovvuri

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Education

Carnegie Mellon University, Master of Science, Computational Data Science Aug 2025 – Dec 2026 | Pittsburgh, PA
Coursework: Search Engines, Large Language Models Applications, Cloud Computing, Machine Learning Systems; GPA: **4.0/4.0**

Shiv Nadar University, Bachelor of Technology, Computer Science Aug 2020 – May 2024 | Delhi, India
Minor: Mathematics, Specialization: Machine Learning, GPA: **9.12/10 (High Distinction)**

Work Experience

AI Engineer Intern, Klaviyo Jun 2026 – Aug 2026 | Boston, MA

- Developed production unsubscribe-risk prediction systems using **PyTorch, Spark, and Ray**, building scalable feature engineering, distributed inference, and evaluation pipelines that achieved a **2.3% reduction in unsubscribe rates**.
- Engineered a **ReAct-style LangGraph agent** leveraging LLMs, expert agent tool-calling workflows, and **Braintrust-based evaluations** to deliver conversational marketing analytics across **4,000+ beta accounts**.

Research Assistant, Catalyst Group - Carnegie Mellon University Feb 2026 – Apr 2026 | Pittsburgh, PA

- Built compiler and inference components across **3 open-source** ML systems stacks: Apache TVM (**C++ ML compiler**), MLC-LLM (**LLM deployment**), WebLLM (**WebGPU browser inference**), improving deployment performance and device compatibility.
- Extended **FlashInfer-Bench**, an **LLM inference benchmarking framework**, to support new model architectures (**Kimi K2.5, GLM-5**) by extracting kernel definitions and tracing real inference workloads from **SGLang**, improving benchmark coverage.
- Built FP16 GEMM kernels for NVIDIA Blackwell GPUs using **TVM/TIRX**, a Python DSL for kernel development, implementing tiling, TMA async loads, pipelining, warp specialization, and cluster optimizations to improve inference efficiency.

Software Engineer, Juspay Technologies Dec 2023 – Jul 2025 | Bangalore, India

- Built CodeGen, an internal **RAG-based** developer tool that grounded LLMs in the existing codebase, automating **28% of payment gateway integration** work and reducing developer effort.
- Integrated **6 payment gateways** into the company's **payment orchestrator**, building and maintaining business-critical payment logic and **encryption workflows** that protected transaction integrity.
- Utilized **Kibana** for transaction log analysis and visualized **Redis** cache performance per API flow through structured logging, enabling faster detection of caching inefficiencies.

Data Science Intern, Code for GovTech 2023 (Open Source Program) Jul 2023 – Aug 2023 | Remote

- Built an on-demand **data generation** and **fine-tuning pipeline** for **Hugging Face models** from natural-language user prompts, automating custom model adaptation workflows.
- Applied Stanford NLP's **Demonstrate-Search-Predict** framework to improve LLM responses on government scheme queries.

Projects

Model-Based Approaches for Data-Effective Pretraining, CX Group, CMU Feb 2026 – May 2026

- Built offline **Common Crawl** crawl simulator with **model based quality scoring**, **seed curation**, and **W&B metrics** aligned to **production pretraining crawls**.

Twitter Recommendation Engine, Cloud Computing, CMU Feb 2026 – Mar 2026

- Architected a **3-tier Go microservice with gRPC**, hitting **1,000+ RPS** for hundreds of concurrent users under a **\$0.27/hr** budget.
- Built a **Spark ETL pipeline**, processing **1TB (200M+ tweets)** of JSON into a denormalized **MySQL** database.
- Automated fault-tolerant cloud infrastructure and **CI/CD pipelines** natively on AWS using **Terraform, Helm, and Kubernetes**.

Distributed Training Systems, Machine Learning Systems, CMU Jan 2026 – Feb 2026

- Implemented **2D parallel training (Data + Tensor Model Parallelism)** from scratch using **MPI collectives** (AllReduce, AllGather, Reduce-Scatter), including **Megatron-style** tensor parallel forward/backward communication.
- Built **ZeRO Stage-3 parameter sharding** with shard-local optimizer states, enabling memory-efficient distributed training.

QryEval, Search Engines, CMU Sep 2025 – Nov 2025

- Built an end-to-end **Neural Search & RAG Engine** utilizing **BM25** and dense vector retrieval, optimizing query relevance (MAP/NDCG) through **Learning-to-Rank (LTR)** pipelines powered by **BERT** and pseudo-relevance feedback algorithms.

Publications

Reddy, K.S.G., Prabhakar, M. and Mukherjee, S., 2025, March. **Revisiting Subject-Action Relevance for Egocentric Activity Recognition**. In *2025 National Conference on Communications (NCC)* (pp. 1-6). IEEE. [↗](#)

Reddy, K.S.G., Bodduluri, S., Adityaja, A.M., Shigwan, S., Kumar, N., Mukherjee, S., 2024, November. **UnSeGArmaNet: Unsupervised Image Segmentation using Graph Neural Networks with Convolutional ARMA Filters**. In *2024 British Machine Vision Conference (BMVC'24), Glasgow, UK*. [↗](#)

Skills

Programming Languages: Python, Go, C++

Libraries & Frameworks: PyTorch, LangGraph FastMCP, scikit-learn, Hugging Face, NumPy, Pandas, FastAPI, FAISS, vLLM

Cloud & Databases: AWS, Azure, GCP, MySQL, PostgreSQL, MongoDB, Redis

Developer Tools & Infrastructure: Git, Docker, Kibana, Terraform, Helm, Kubernetes